
$$\mathcal{M}\mathbb{R}^4$$

$$VV^\mu\omega_\mu T_{\mu\nu}T^{\mu\nu}$$

$$g_{\mu\nu}(-,+,+,+)(+,-,-,-)$$

$$s^2=g_{\mu\nu}\,x^\mu x^\nu\,.$$

$$v^\mu(-,+,+,+)$$

$$g_{\mu\nu}v^\mu v^\nu < 0 \quad ,$$

$$g_{\mu\nu}v^\mu v^\nu = 0 \quad ,$$

$$g_{\mu\nu}v^\mu v^\nu > 0 \quad .$$

$$x^\mu(\lambda)$$

$$\frac{{}^2x^\mu}{\lambda^2}+\Gamma^\mu_{\alpha\beta}\frac{x^\alpha}{\lambda}\frac{x^\beta}{\lambda}=0.$$

$$\Gamma^\mu_{\alpha\beta}=\frac{1}{2}g^{\mu\nu}\left(\partial_\alpha g_{\nu\beta}+\partial_\beta g_{\nu\alpha}-\partial_\nu g_{\alpha\beta}\right).$$

$$R^\rho{}_{\sigma\mu\nu}=\partial_\mu\Gamma^\rho_{\nu\sigma}-\partial_\nu\Gamma^\rho_{\mu\sigma}+\Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma}-\Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}.$$

$$R_{\mu\nu}=R^\rho{}_{\mu\rho\nu},$$

$$R=g^{\mu\nu}R_{\mu\nu}.$$

$$G_{\mu\nu}=R_{\mu\nu}-\frac{1}{2}Rg_{\mu\nu}.$$

$$\nabla^\mu G_{\mu\nu}=0.$$

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

$$S[\phi]=\int \mathcal{L}(\phi,\partial\phi,x)^4x.$$

$$\delta S=0.$$

$$S_G=\frac{c^3}{16\pi G}\int R\sqrt{-g}^4x.$$

$$g=(g_{\mu\nu})$$

$$S=S_G+S_M.$$

$$T_{\mu\nu}=-\frac{2}{\sqrt{-g}}\frac{\delta S_M}{\delta g^{\mu\nu}}.$$

$$T_{\mu\nu}$$

$$\square$$

$$|\psi\rangle\mathcal{H}$$

$$|a\rangle|b\rangle$$

$$\alpha,\beta\in\mathbb{C}$$

$$|\psi\rangle=\alpha\,|a\rangle+\beta\,|b\rangle$$

$$|\alpha|^2+|\beta|^2=1.$$

$$A$$

$$A=A^\dagger.$$

$$|\psi\rangle$$

$$\langle A \rangle_\psi = \langle \psi | \, A \, | \psi \rangle \, .$$

$$A|a_i\rangle a_i|\left\langle a_i|\psi\right\rangle|^2a_i$$

$$\phi(x)\hat{\phi}(x)$$

$$T_{\mu\nu}(x)\longrightarrow \hat{T}_{\mu\nu}(x).$$

$$|L\rangle |R\rangle$$

$$|\psi\rangle=\frac{1}{\sqrt{2}}(|L\rangle+|R\rangle).$$

$$T_{\mu\nu}\hat{T}_{\mu\nu}$$

$$G_{\mu\nu}=\frac{8\pi G}{c^4}\langle\psi|\hat{T}_{\mu\nu}|\psi\rangle.$$

$$G_{\mu\nu}[g] = \frac{8\pi G}{c^4} \langle \hat{T}_{\mu\nu} \rangle.$$

□

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu},$$

$$\eta_{\mu\nu}h_{\mu\nu}$$

$$h_{\mu\nu}$$

$$h_{\mu\nu} \mapsto h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu.$$

$$h_{\mu\nu}$$

$$\Box h_{\mu\nu} = 0$$

$$h_{\mu\nu}$$

$$\square$$

$$h_{\mu\nu}$$

$$\mathcal{L} = \kappa h_{\mu\nu} T^{\mu\nu}.$$

$$\kappa$$

□

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|L\rangle + |R\rangle).$$

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|L\rangle |g_L\rangle + |R\rangle |g_R\rangle).$$

□

$$h_{\mu\nu}$$

$$\hat{h}_{\mu\nu}(x)=\sum_{\lambda}\int\frac{{}^3k}{(2\pi)^3}\mathcal{N}_k\left(\epsilon_{\mu\nu}^{(\lambda)}(k)\hat{a}_{k,\lambda}e^{-ikx}+\epsilon_{\mu\nu}^{(\lambda)*}(k)\hat{a}_{k,\lambda}^{\dagger}e^{ikx}\right).$$

$$S=\int\sqrt{-g}\left(\frac{c^3}{16\pi G}R+c_1R^2+c_2R_{\mu\nu}R^{\mu\nu}+\cdots\right)^4x.$$

$$h_{\mu\nu}T_{\mu\nu}h_{\mu\nu}h_{\mu\nu}$$

$$\square$$

$$\mathcal{G}_C$$

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}.$$

$$t_P = \sqrt{\frac{\hbar G}{c^5}}.$$

$$m_P = \sqrt{\frac{\hbar c}{G}}.$$

$$r_s = \frac{2GM}{c^2}.$$

$$S_{BH} = \frac{k_B c^3 A}{4G\hbar}.$$

$$G\hbar$$

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}.$$

$$\hbar$$

$$\square$$

$$\Rightarrow \hat{T}_{\mu\nu}$$

$$\hat{T}_{\mu\nu}g_{\mu\nu}$$

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

$$h_{\mu\nu}$$

$$\Rightarrow \hat{h}_{\mu\nu}$$

$$\Rightarrow$$

$$|\psi\rangle=\frac{1}{\sqrt{2}}(|L\rangle+|R\rangle).$$

$$g_{\mu\nu}=\eta_{\mu\nu}+h_{\mu\nu}$$

$$h_{\mu\nu}$$

$$T_{\mu\nu}$$

$$\ell_P=\sqrt{\frac{\hbar G}{c^3}}$$

$$hGc$$

$$[\hbar] = ML^2T^{-1}, \quad [G] = M^{-1}L^3T^{-2}, \quad [c] = LT^{-1}$$

$$\frac{hG}{c^3} \sim L^2.$$

2mm]Ingolf Lohmann